# TIØ4145 Corporate Finance Chapters 20,21

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Monday, October 27, 2025

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## Primary fields of interests

Microeconomics, industrial economics and quantitative finance

## Main research interests

Real options, dynamic (investment) decision making under uncertainty

## Academic background

| | |
|---|---|
| 2008 - 2011 | PhD in Economics |
| 2003 - 2008 | Dipl.-Ing. (5-year degree) Technical Mathematics, Technische Universität Graz, Austria |
| 2006 - 2007 | MSc Quantitative Finance and Actuarial Sciences, Tilburg University, The Netherlands |

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# Schedule

| Week | Subject | Lecturer | Chapter |
|---|---|---|---|
| 44 | Financial options, Option valuation | Verena | 20,21 |
| 45 | Option valuation, Real options | Verena | 21, 22 |

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# Financial Options

- Chapter 20
  - 20.1 Option Basics
  - 20.2 Option Payoffs at Expiration
  - 20.3 Put-Call Parity
  - 20.4 Factors Affecting Option Prices
  - ~~20.5 Exercising Options Early~~
  - ~~20.6 Options and Corporate Finance~~

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# Learning objectives

- Define the following terms: **call option, put option, exercise price, strike price, exercising the option, expiration date, American option, European option, in-the-money**, and **out-of-the-money**.
- Compute the value of a call or a put option at expiration.
- Use **put-call parity** to solve for the call premium, the put premium, the stock price, the strike price, or the dividend.
- Discuss the following **factors that influence call and put option values**: stock price, strike price, and volatility.

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# Option Basics

- Financial Option
  - A contract that gives its owner the right (but not the obligation) to purchase or sell an asset at a fixed price at some future date
  - Two distinct kinds of options
- Call Option
  - A financial option that gives its owner the right to buy an asset
- Put Option
  - A financial option that gives its owner the right to sell an asset
- Option Writer
  - The seller of an option contract
- Stock option
  - Gives holder the option to buy or sell a share of stock on or before a given date for a given price.

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# Understanding Option Contracts

- Exercising an Option
  - When a holder of an option enforces the agreement and buys or sells a share of stock at the agreed-upon price
- Strike Price (Exercise Price)
  - The price at which an option holder buys or sells a share of stock when the option is exercised
- Expiration Date
  - The last date on which an option holder has the right to exercise the option

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# Understanding Option Contracts

- American Option
  - Options that allow their holders to exercise the option on any date up to, and including, the expiration date
- European Option
  - Options that allow their holders to exercise the option only on the expiration date
- Vanilla Options vs Exotic options or Binary options

*Note: The names American and European have nothing to do with the location where the options are traded.*

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# Understanding Option Contracts

- The option buyer (holder)
  - Holds the right to exercise the option and has a long position in the contract
- The option seller (writer)
  - Sells (or writes) the option and has a short position in the contract
  - Because the long side has the option to exercise, the short side has an obligation to fulfill the contract if it is exercised.
- The buyer pays the writer a premium.
  - Market price of option = option premium

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# Call Option

- Link https://www.youtube.com/watch?v=EfmTWu2yn5Q
- Start 00:26

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# Interpreting Stock Option Quotations

- Stock options are traded on organized exchanges.
- By convention, all *standard listed equity options* expire on the Saturday following the third Friday of the month.
- Open Interest
  - The total number of contracts of a particular option that have been written

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# Options Basics

- At-the-money
  - Describes an option whose exercise price is equal to the current stock price
- In-the-money
  - Describes an option whose value if immediately exercised would be positive
- Out-of-the-money
  - Describes an option whose value if immediately exercised would be negative

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# Options Basics

- Deep In-the-money
  - Describes an option that is in-the-money and for which the strike price and the stock price are very far apart
- Deep Out-of-the-money
  - Describes an option that is out-of–the-money and for which the strike price and the stock price are very far apart

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## eBay Inc (ebay)

2025-10-24 15:59:59 ET (Delayed) Bid: 97 Ask: 98 Vol: 4,352,009 Last: 97 Change: +1.68 (+1.7284%)

**Filters By:**

| Volume: | Expiration Type: | Options Range: | Size: | Expiration: | |
|---|---|---|---|---|---|
| All | All | Near the Money | 3 | 2025 November | View Chain |

## Options Chain

Total Records: 28

### Fri Nov 07 2025

| Calls | | | | | | | | | Strike | Puts | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int | | Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int |
| 6.08 | +1.08 | 5.5 | 6.3 | 4 | 0.53 | 0.6604 | 0.0391 | 32 | EBAY 94.000 | 2.17 | -0.78 | 1.96 | 2.27 | 21 | 0.47 | -0.3412 | 0.0394 | 54 |
| 5.25 | +0.8 | 4 | 5.2 | 11 | 0.44 | 0.6197 | 0.0409 | 68 | EBAY 95.000 | 2.64 | -0.735 | 2.49 | 2.66 | 12 | 0.48 | -0.3821 | 0.0412 | 36 |
| 5.2 | +1.3 | 4.3 | 4.45 | 2 | 0.48 | 0.5778 | 0.0422 | 70 | EBAY 96.000 | 3.1 | -0.775 | 2.77 | 3.1 | 13 | 0.47 | -0.4244 | 0.0425 | 11 |
| 3.9 | +0.45 | 3.65 | 3.95 | 65 | 0.47 | 0.535 | 0.043 | 39 | EBAY 97.000 | 3.35 | -1.075 | 3.25 | 3.6 | 25 | 0.47 | -0.4675 | 0.0433 | 21 |
| 2.66 | +0 | 3.2 | 3.45 | 0 | 0.48 | 0.492 | 0.0433 | 17 | EBAY 98.000 | 5.3 | +0 | 3.85 | 4.1 | 0 | 0.48 | -0.5108 | 0.0436 | 13 |
| 3.1 | +0.425 | 1.17 | 3.15 | 30 | 0.38 | 0.4494 | 0.043 | 19 | EBAY 99.000 | 0 | +0 | 4.4 | 4.7 | 0 | 0.48 | -0.5538 | 0.0434 | 0 |
| 2.8 | +0.525 | 2.5 | 2.63 | 22 | 0.48 | 0.4076 | 0.0423 | 121 | EBAY 100.000 | 7.15 | +0 | 5 | 5.3 | 0 | 0.47 | -0.596 | 0.0427 | 1 |

### Fri Nov 14 2025

| Calls | | | | | | | | | Strike | Puts | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int | | Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int |
| 5.4 | +0 | 4 | 7.95 | 0 | 0.43 | 0.6571 | 0.0359 | 12 | EBAY 94.000 | 0 | +0 | 2.3 | 4.6 | 0 | 0.55 | -0.3466 | 0.0363 | 0 |
| 5.74 | +0.94 | 4.65 | 7 | 2 | 0.48 | 0.62 | 0.0373 | 26 | EBAY 95.000 | 2.84 | -0.81 | 2.71 | 3 | 114 | 0.43 | -0.3844 | 0.0378 | 2 |
| 3.86 | +0 | 4.6 | 4.9 | 0 | 0.43 | 0.5818 | 0.0385 | 24 | EBAY 96.000 | 3.3 | -0.85 | 3.1 | 3.45 | 1 | 0.43 | -0.4228 | 0.0389 | 0 |
| 4.53 | +0.705 | 4.05 | 4.4 | 2 | 0.42 | 0.5428 | 0.0392 | 6 | EBAY 97.000 | 4.61 | +0 | 3.55 | 3.95 | 0 | 0.43 | -0.4623 | 0.0397 | 2 |
| 3.98 | +0.53 | 3.35 | 3.9 | 2 | 0.41 | 0.5036 | 0.0396 | 10 | EBAY 98.000 | 4.1 | -1.1 | 4.05 | 4.45 | 11 | 0.43 | -0.5025 | 0.0402 | 2 |
| 3.3 | +0.3 | 3.15 | 3.45 | 2 | 0.42 | 0.4649 | 0.0395 | 2 | EBAY 99.000 | 4.75 | -1.1 | 4.65 | 5 | 11 | 0.43 | -0.5422 | 0.0402 | 0 |
| 2.83 | +0.23 | 2.71 | 3.05 | 19 | 0.42 | 0.4264 | 0.0391 | 50 | EBAY 100.000 | 5.28 | -1.22 | 5.25 | 5.6 | 5 | 0.43 | -0.5814 | 0.0398 | 0 |

### Fri Nov 21 2025

| Calls | | | | | | | | | Strike | Puts | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int | | Last | Net | Bid | Ask | Vol | IV | Delta | Gamma | Int |
| 6.89 | +1.265 | 4.55 | 6.8 | 1 | 0.35 | 0.6496 | 0.0338 | 36 | EBAY 94.000 | 2.66 | -0.79 | 2.64 | 2.85 | 11 | 0.4 | -0.3531 | 0.0342 | 136 |
| 5.7 | +0.6 | 5.45 | 5.8 | 57 | 0.4 | 0.6146 | 0.035 | 1342 | EBAY 95.000 | 3 | -0.9 | 3 | 3.25 | 11 | 0.4 | -0.3885 | 0.0354 | 460 |
| 5.45 | +0.875 | 3.25 | 5.3 | 6 | 0.32 | 0.5788 | 0.036 | 415 | EBAY 96.000 | 3.45 | -0.95 | 3.4 | 3.7 | 2 | 0.4 | -0.4248 | 0.0364 | 44 |
| 4.75 | +0.625 | 4.45 | 4.7 | 3 | 0.4 | 0.5424 | 0.0366 | 39 | EBAY 97.000 | 4.1 | -0.85 | 3.9 | 4.2 | 18 | 0.4 | -0.4617 | 0.037 | 45 |
| 4.35 | +0.5 | 4.3 | 4.45 | 173 | 0.4 | 0.5241 | 0.0368 | 1871 | EBAY 97.500 | 4.15 | -1 | 4.15 | 4.45 | 18 | 0.4 | -0.4802 | 0.0372 | 275 |
| 4.25 | +0.6 | 3.35 | 4.25 | 11 | 0.37 | 0.5058 | 0.0368 | 33 | EBAY 98.000 | 4.34 | -1.035 | 4.4 | 4.7 | 32 | 0.4 | -0.4987 | 0.0374 | 63 |
| 4.18 | +0.93 | 2.03 | 3.85 | 6 | 0.33 | 0.4694 | 0.0368 | 19 | EBAY 99.000 | 4.85 | -1.225 | 4.95 | 5.25 | 3 | 0.4 | -0.5356 | 0.0374 | 38 |

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# EQUINOR - Stock Option

OSLO BØRS DERIVATIVES
STOCK OPTION

| EQN | Volume 24/10/2023 09:59 | Open Interest 23/10/2023 | Currency | Time zone |
|---|---|---|---|---|
| | - | 5,360 | NOK | CET |

DELAYED PRICES | SPECIFICATION | SETTLEMENT PRICES

## UNDERLYING

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Name | Equinor ASA | ISIN | NO0010096985 | | Market | Oslo Børs | |
| Currency | NOK | Best Bid | 373.65 | 24 Oct 23 07:44 | Best Ask | 373.80 | 24 Oct 23 07:44 |
| Time | CET | Last | 373.65 | 24 Oct 23 07:44 | Last change | 0.38 % | |
| Volume | 221,871 | High | 373.65 | | Low | 368.55 | |

Expiry

Nov 2023 ×

Select all | Compact | All strikes | SUBMIT

## NOVEMBER 2023 PRICES - 24/10/23

| SETTL. | LAST | BID | ASK | | STRIKE | | BID | ASK | LAST | SETTL. |
|---|---|---|---|---|---|---|---|---|---|---|
| 26.85 | - | 26.75 | 29.00 | C | 350.00 | P | 4.00 | 4.60 | - | 4.75 |
| 22.96 | - | 23.00 | 24.75 | C | 355.00 | P | 5.20 | 5.70 | - | 6.00 |
| 19.31 | - | 19.50 | 20.50 | C | 360.00 | P | 6.80 | 7.10 | - | 7.57 |
| 15.92 | - | 16.25 | 17.25 | C | 365.00 | P | 8.50 | 9.00 | - | 9.34 |
| 13.04 | - | 13.25 | 13.75 | C | 370.00 | P | 10.75 | 11.00 | - | 11.59 |
| 10.26 | - | 10.75 | 11.25 | C | 375.00 | P | 13.25 | 13.50 | - | 14.24 |
| 8.21 | - | 8.25 | 9.00 | C | 380.00 | P | 16.00 | 16.75 | - | 17.27 |
| 6.35 | - | 6.50 | 7.00 | C | 385.00 | P | 19.00 | 20.00 | - | 20.54 |
| 4.78 | - | 4.80 | 5.40 | C | 390.00 | P | 22.25 | 23.75 | - | 24.29 |
| 3.62 | - | 3.55 | 4.20 | C | 395.00 | P | 26.00 | 27.75 | - | 28.06 |
| 2.67 | - | 2.45 | 3.20 | C | 400.00 | P | 29.75 | 32.00 | - | 32.21 |

SE



https://live.euronext.com/en/product/stock-options/EQN-DOSL

# Options on Other Financial Securities

- Although the most commonly traded options are on stocks, options on other financial assets, like the S&P 100 index, the S&P 500 index, the Dow Jones Industrial index, and the NYSE index, are also traded.
- Options are also traded on: treasury securities, currencies and commodities.

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# Options Basics

- Hedge
  - To reduce risk by holding contracts or securities whose payoffs are negatively correlated with some risk exposure
- Speculate
  - When investors use contracts or securities to place a bet on the direction in which they believe the market is likely to move

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# Option Payoffs at Expiration

- Long Position in an Option Contract
  - The value of a **call option** at expiration is

$$C = \max(S - K, 0) \qquad (20.1)$$

  - Where *S* is the stock price at expiration, *K* is the exercise price, *C* is the value of the call option, and max is the maximum of the two quantities in the parentheses

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# Payoff of a Call Option with a Strike Price of $20 at Expiration

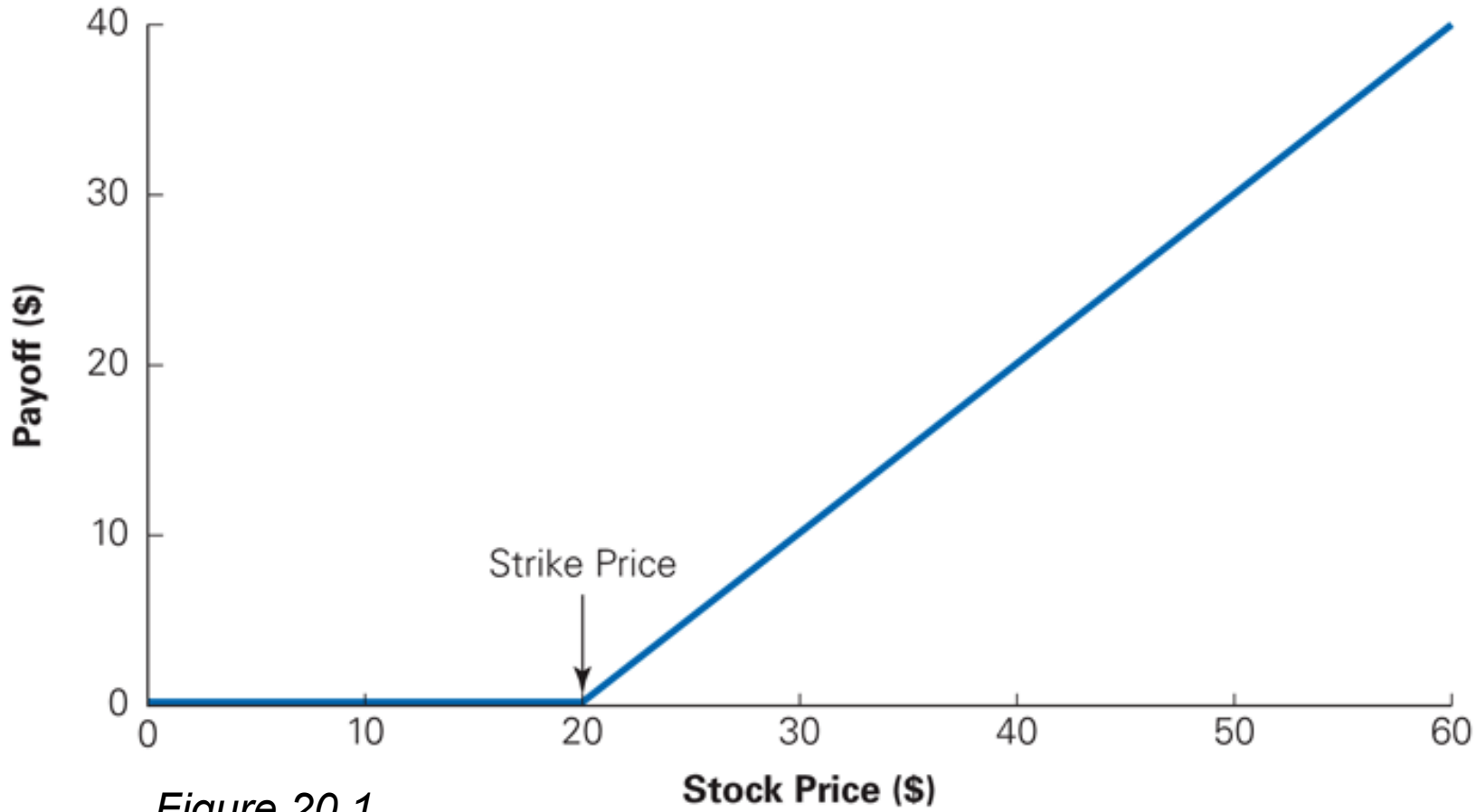


*Figure 20.1*

# Option Payoffs at Expiration

- Long Position in an Option Contract
  - The value of a **put option** at expiration is

$$P = \max(K - S, 0) \qquad (20.2)$$

  - Where *S* is the stock price at expiration, *K* is the exercise price, *P* is the value of the put option, and max is the maximum of the two quantities in the parentheses

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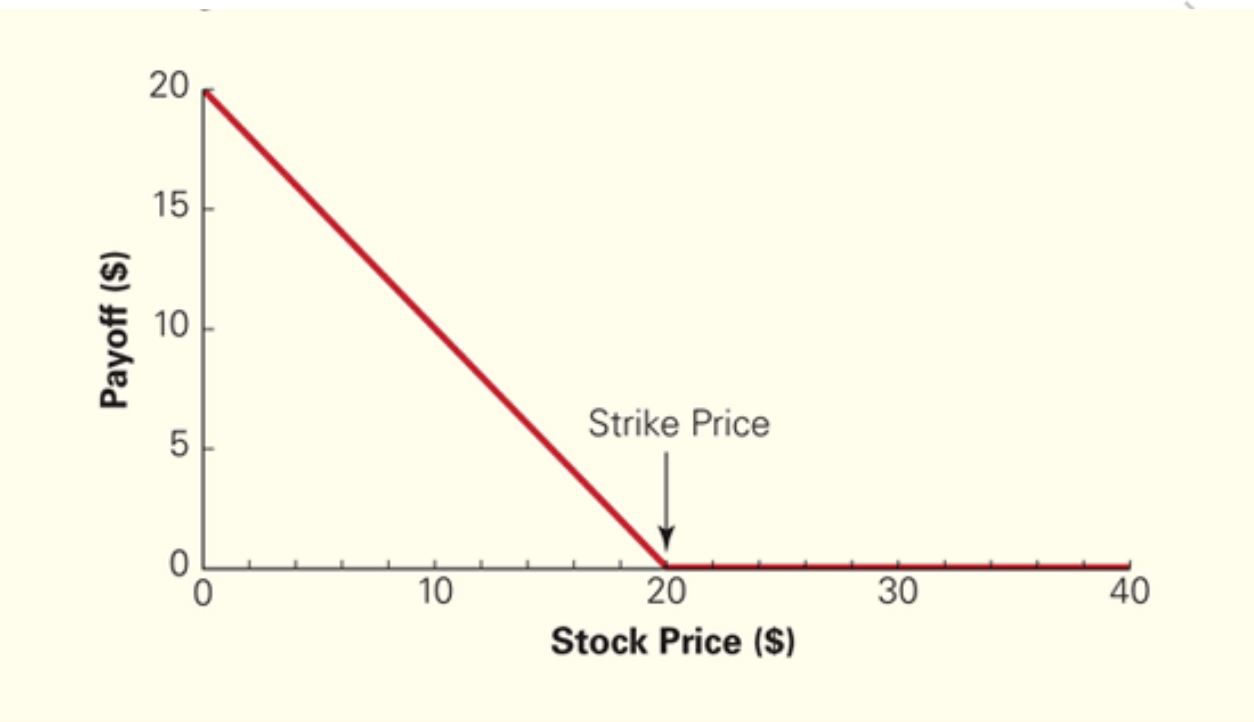
# Payoff of a put option at maturity




*Example 20.2*

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# Short Position

- Short Position in an Option Contract
  - An investor that sells an option has an obligation.
    - This investor takes the opposite side of the contract to the investor who bought the option.
    - Thus the seller's cash flows are the negative of the buyer's cash flows.

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# Short Position in a Call Option at Expiration

*Figure 20.2*

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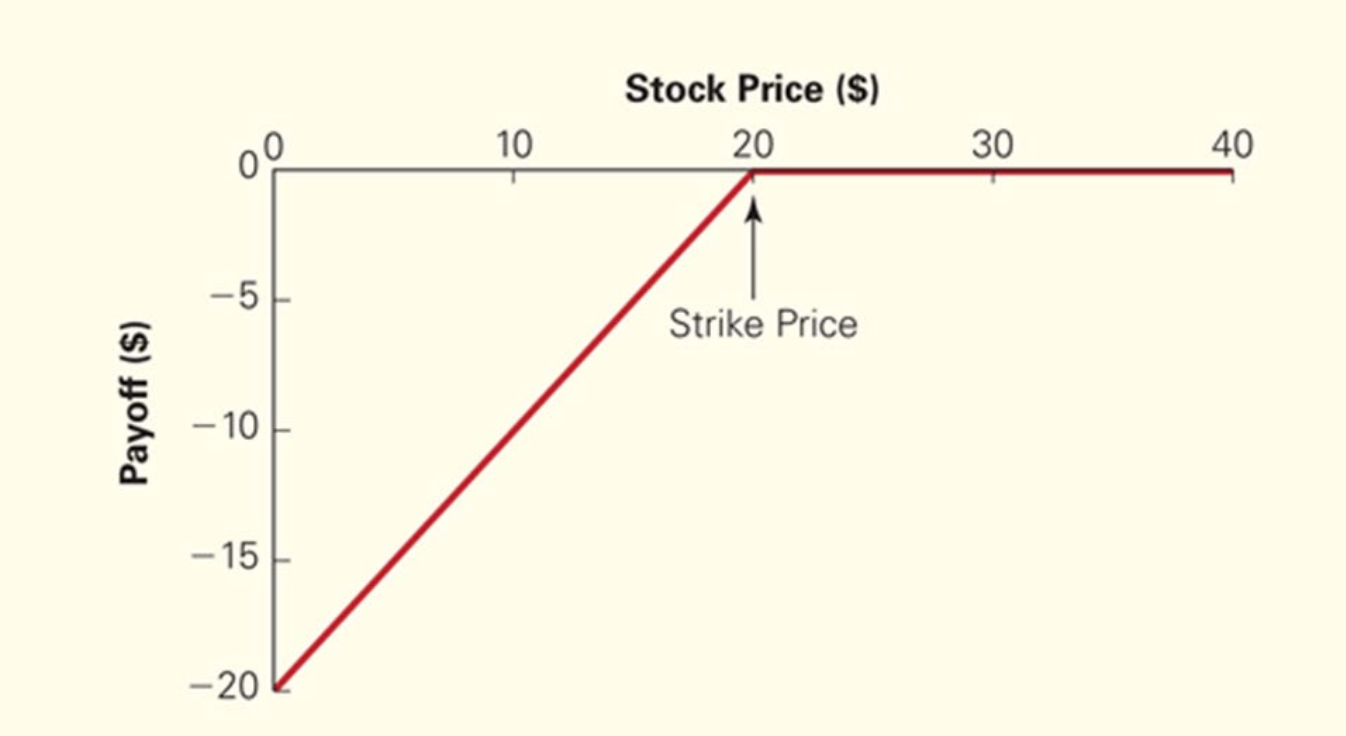
# Payoff of a short position in a put option at maturity




*Example 20.3*

# Profits for Holding an Option to Expiration

- Although payouts on a long position in an option contract are never negative, the profit from purchasing an option and holding it to expiration could be negative because the payout at expiration might be less than the initial cost of the option.

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# Example (also see example 20.4)

## Profit on Holding a Position in a Put Option until Expiration

### Problem

Assume you decided to purchase each of the August put options quoted in Table 20.1 on July 8, 2009, and you financed each position by shorting a two-month bond with a yield of 3%. Plot the profit of each position as a function of the stock price on expiration.

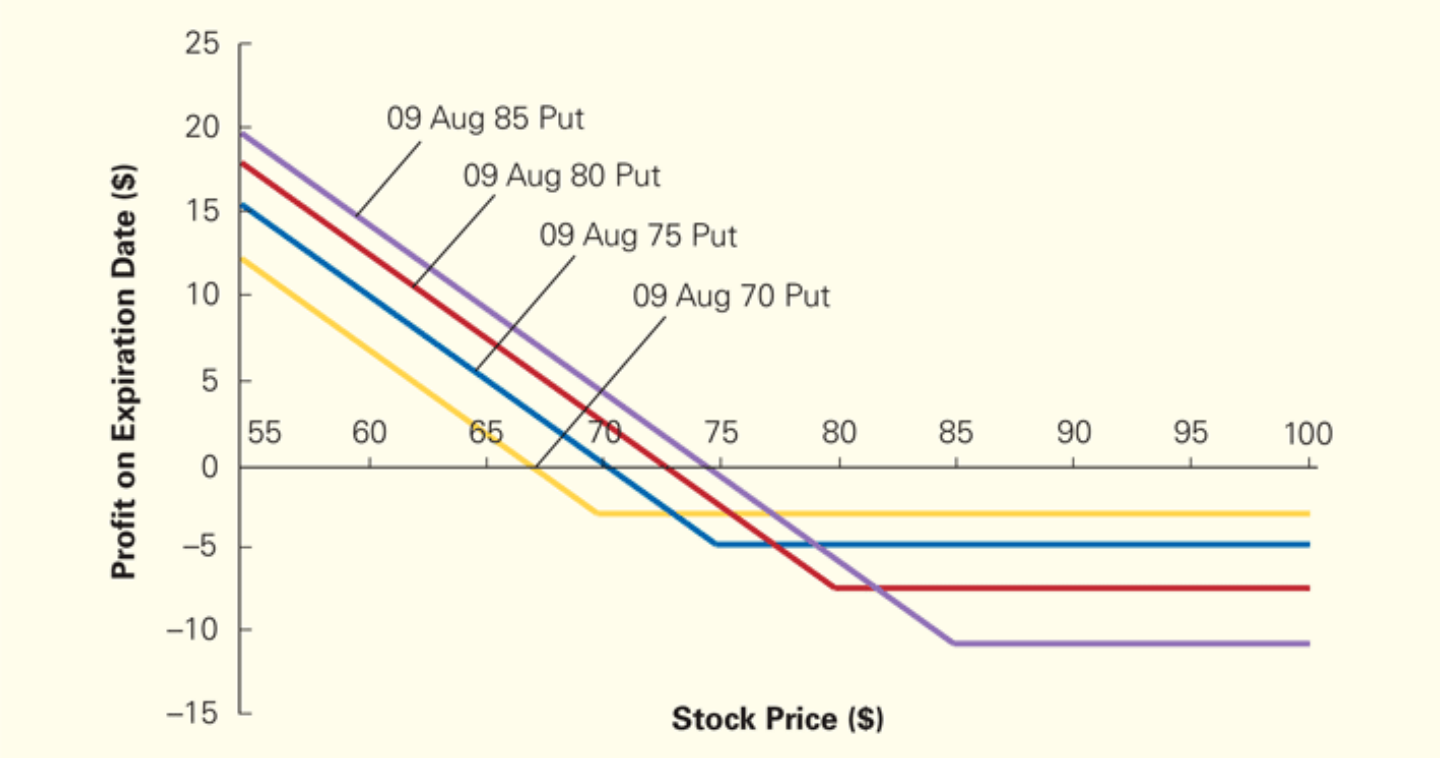
**AMZN** **77.03** **+1.40**

Jul 08 2009 @ 15:26 ET **Bid** 77.02 **Ask** 77.03 **Size** 1 x 3 **Vol** 6548487

| Calls | Last Sale | Net | Bid | Ask | Vol | Open Int | Puts | Last Sale | Net | Bid | Ask | Vol | Open Int |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 09 Jul 70.00 (ZQN GN-E) | 7.65 | 1.60 | 7.20 | 7.30 | 221 | 2637 | 09 Jul 70.00 (ZQN SN-E) | 0.36 | –0.18 | 0.36 | 0.38 | 684 | 11031 |
| 09 Jul 75.00 (ZQN GO-E) | 3.35 | 0.86 | 3.20 | 3.30 | 943 | 6883 | 09 Jul 75.00 (ZQN SO-E) | 1.30 | –0.66 | 1.38 | 1.40 | 2394 | 15545 |
| 09 Jul 80.00 (QZN GP-E) | 0.94 | 0.24 | 0.93 | 0.96 | 2456 | 9877 | 09 Jul 80.00 (QZN SP-E) | 4.15 | –1.05 | 4.00 | 4.10 | 700 | 10718 |
| 09 Jul 85.00 (QZN GQ-E) | 0.22 | 0.07 | 0.19 | 0.21 | 497 | 26679 | 09 Jul 85.00 (QZN SQ-E) | 8.25 | –1.25 | 8.25 | 8.35 | 112 | 7215 |
| 09 Aug 70.00 (ZQN HN-E) | 9.75 | 1.04 | 9.60 | 9.70 | 51 | 326 | 09 Aug 70.00 (ZQN TN-E) | 2.77 | –0.39 | 2.75 | 2.79 | 225 | 1979 |
| 09 Aug 75.00 (ZQN HO-E) | 6.50 | 0.70 | 6.40 | 6.50 | 65 | 1108 | 09 Aug 75.00 (ZQN TO-E) | 4.60 | –0.55 | 4.55 | 4.60 | 2322 | 6832 |
| 09 Aug 80.00 (QZN HP-E) | 4.00 | 0.50 | 3.90 | 4.00 | 172 | 2462 | 09 Aug 80.00 (QZN TP-E) | 6.95 | –0.95 | 7.05 | 7.15 | 145 | 2335 |
| 09 Aug 85.00 (QZN HQ-E) | 2.15 | 0.15 | 2.22 | 2.26 | 833 | 5399 | 09 Aug 85.00 (QZN TQ-E) | 10.15 | –1.00 | 10.30 | 10.40 | 43 | 4599 |

*Source*: Chicago Board Options Exchange at www.cboe.com

# Example (also see example 20.4)

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# Option Returns from Purchasing an Option and Holding It to Expiration

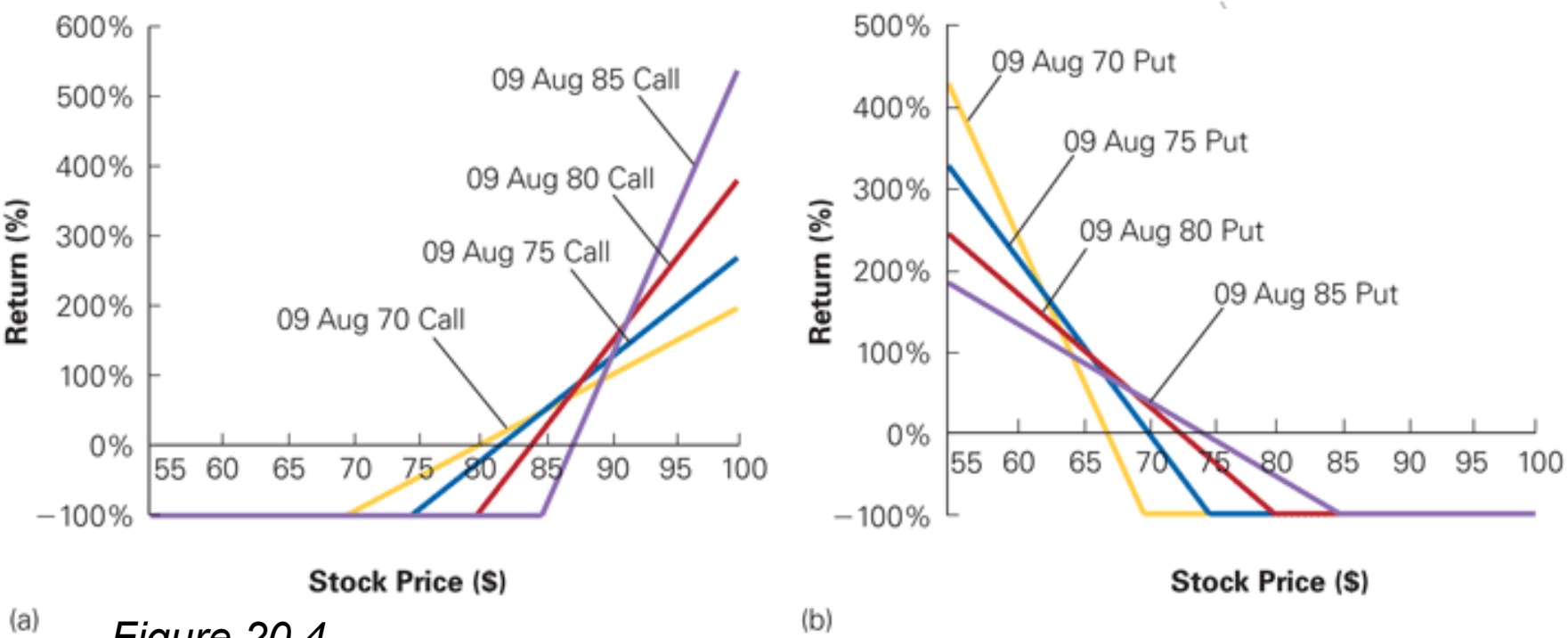


*Figure 20.4*

(a) The <u>return on the expiration date</u> from purchasing one of the August call options in Table 20.1 on July 8, 2009, and holding the position until the expiration date; (b) the same return for the August put options in the table.

# Returns for Holding an Option to Expiration

- The maximum loss on a purchased **call option** is 100% (when the option expires worthless).
- Out-of-the money call options are more likely to expire worthless, but if the stock goes up sufficiently it will also have a much higher return than an in-the-money call option.
- Call options have more extreme returns than the stock itself.

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# Returns for Holding an Option to Expiration

- The maximum loss on a purchased **put option** is 100% (when the option expires worthless).
- Put options will have higher returns in states with low stock prices.
- Put options are generally not held as an investment, but rather as insurance to hedge other risk in a portfolio.

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# Combinations of Options

- Straddle
  - A portfolio that is long a call option and a put option on the same stock with the same exercise date and strike price
    - May be used if investors expect the stock to be very volatile and move up or down a large amount, but do not necessarily know which direction the stock will move.

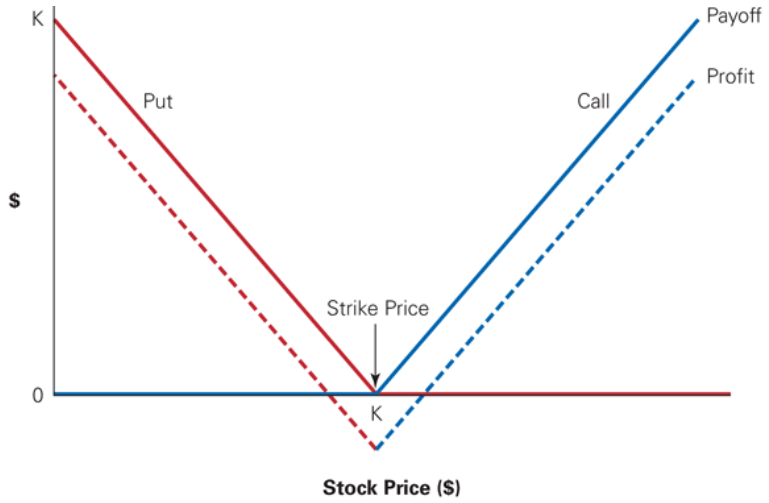


*Figure 20.5*

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# Combinations of Options

- Strangle
  - A portfolio that is long a call option and a put option on the same stock with the same exercise date but the strike price on the call exceeds the strike price on the put

*Figure 20.5*

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# Combinations of Options

- Butterfly Spread
  - A portfolio that is long two call options with differing strike prices, and short two call options with a strike price equal to the average strike price of the first two calls (https://www.youtube.com/watch?v=vKATKsDfxU8)
    - A butterfly spread makes money when the stock and strike prices are close.

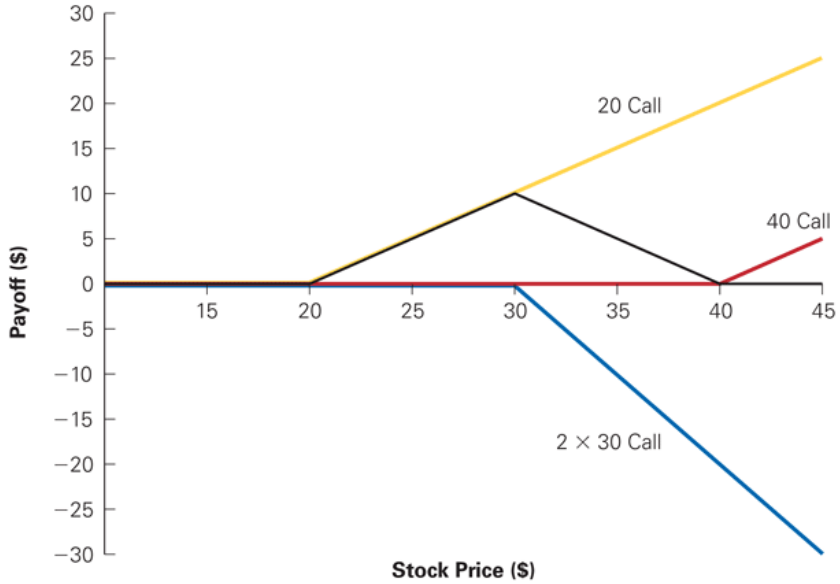


*Figure 20.5*

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# Combinations of Options (cont'd)

- Protective Put
  - A long position in a put held on a stock you already own.
- Portfolio Insurance
  - A protective put written on a portfolio rather than a single stock. When the put does not itself trade, it is synthetically created by constructing a replicating portfolio
  - Portfolio insurance can also be achieved by purchasing a bond and a call option.

# Portfolio Insurance

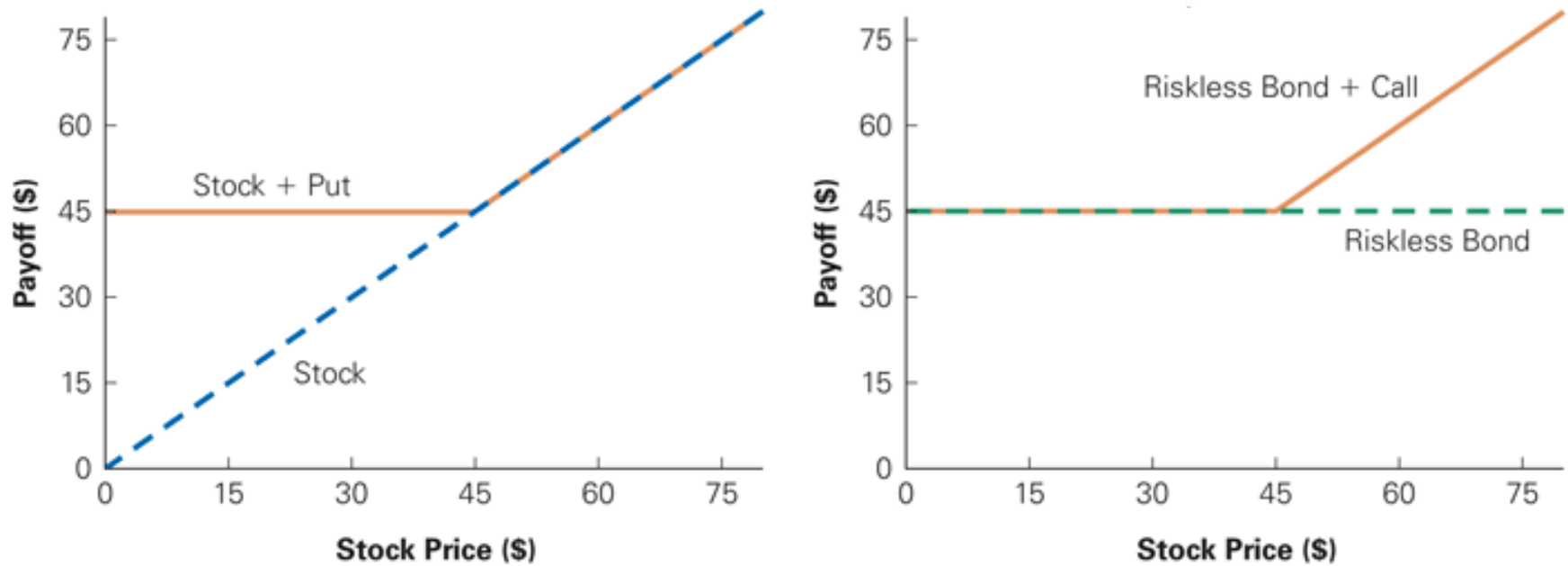


*Figure 20.5*

The plots show two different ways to <u>insure against the possibility of the price of Amazon stock falling below $45</u>.

The orange line in (a) indicates the value on the expiration date of a position that is long one share of Amazon stock and one European put option with a strike of $45 (the blue dashed line is the payoff of the stock itself).

The orange line in (b) shows the value on the expiration date of a position that is long a zero-coupon risk-free bond with a face value of $45 and a European call option on Amazon with a strike price of $45 (the green dashed line is the bond payoff).

# Put-Call Parity

- Consider the two different ways to construct portfolio insurance discussed above.
  - Purchase the stock and a put
  - Purchase a bond and a call
- Because both positions provide exactly the same payoff, the Law of One Price requires that they must have the same price.

$$S + P = PV(K) + C$$

  - Where K is the strike price of the option (the price you want to ensure that the stock will not drop below), C is the call price, P is the put price, and S is the stock price

# Put-Call Parity

- Rearranging the terms gives an expression for the price of a European call option for a non-dividend-paying stock.

$$C = P + S - PV(K) \qquad (20.3)$$

  - This relationship between the value of the stock, the bond, and call and put options is known as **put-call parity.**

- If the stock pays a dividend, put-call parity becomes

$$C = P + S - PV(K) - PV(Div)$$

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# Factors Affecting Option Prices

- Arbitrage Bounds on Option Prices
  - An American option cannot be worth less than its European counterpart
    - Note that they may have equal value
  - A put option cannot be worth more than its strike price.
  - A call option cannot be worth more than the stock itself.

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# Arbitrage Bounds on Option Prices

- Intrinsic Value
  - Value that an option would have if it expired immediately.
  - The amount by which an option is in-the-money, or zero if the option is out-of-the-money
    - An American option cannot be worth less than its intrinsic value
- Time Value
  - The difference between an option's price and its intrinsic value
    - An American option cannot have a negative time value.

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# Arbitrage Bounds on Option Prices

- Option Prices and Volatility
  - The value of an option generally increases with the volatility of the stock.
  - Intuition: an increase in volatility increases the likelihood of very high and very low returns for the stock.

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# Summary

- A **call option** gives the holder the right (but not the obligation) to purchase an asset at some future date at the strike price. A **put option** gives the holder the right to sell an asset at some future date at the strike price.
- **An American option** can be exercised on any date up to, and including, the exercise date**. A European option** can be exercised only on the expiration date.
- If the payoff from exercising the option immediately is positive, the option is **in-the-money**. If the stock price equals the strike price, the option is **at-the-money**. Finally, if you would lose money by exercising an option immediately, it is **out-of-the-money**.
- The **value of** a call **option at expiration** is $C = max\ (S - K, 0)$. The value of a put option at expiration is $P = max\ (K - S, 0)$
- The value an option would have if it expired today is its **intrinsic value**. The **time value** of an option is the difference between its current value and its intrinsic value.
- **Put-call parity** relates the value of the European call to the European put and the stock. $C = P + S - PV(K) - PV(Div)$
- **Arbitrage bounds for option prices**:
  - An American option cannot be worth less than its European counterpart.
  - A put option cannot be worth more than its exercise price.
  - A call option cannot be worth more than the stock itself.
  - An American option cannot be worth less than its intrinsic value.
  - An American option with a later exercise date cannot be worth less than an otherwise identical American option with an earlier exercise date
- The **value of an option generally increases with the volatility** of the stock.

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# Option Valuation

- Chapter 21
  - **21.1 The Binomial Option Pricing Model (today)**
  - 21.2 The Black-Scholes Option Pricing Model
  - 21.3 Risk-Neutral Probabilities
  - 21.4 Risk and Return of an Option
  - 21.5 Beta of Risky Debt

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# Learning objectives

- Illustrate the use of the **Binomial Option Pricing Model** to value an option.
- Define **the replicating portfolio** for the Binomial Option Pricing Model.
- Use the **Law of One Price** to explain how the Binomial Option Pricing Model provides the correct value under the assumptions made by the model.
- Use the **Black-Scholes Option Pricing formula** to calculate the value of **a call option on a non-dividend-paying stock**.
- Use the **Black-Scholes Option Pricing formula** to calculate the price of a European **put option on a non-dividend paying stock**.
- Compute **the value of a European option on a dividend-paying stock**.
- Define **the Black-Scholes replicating portfolio** for a call option on a non-dividend paying stock or a European put option on a non-dividend paying stock.

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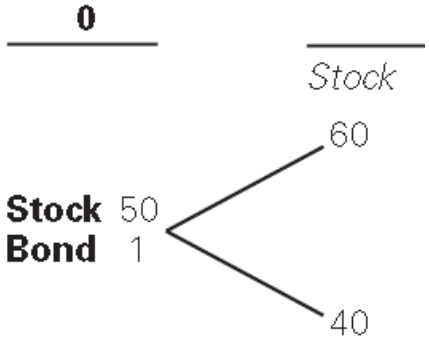
# The Binomial Option Pricing Model

- Binomial Option Pricing Model
  - A technique for pricing options based on the assumption that each period, the stock's return can take on only two values
- Binomial Tree
  - A timeline with two branches at every date representing the possible events that could happen at those times
- Example

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# A Two-State Single-Period Model

- Replicating Portfolio
  - A portfolio consisting of a stock and a risk-free bond that has the same value and payoffs in one period as an option written on the same stock
  - The Law of One Price implies that the current value of the call and the replicating portfolio must be equal.

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# A Two-State Single-Period Model

- Assume
  - A European call option expires in one period and has an exercise price of $50.
  - The stock price today is equal to $50 and the stock pays no dividends.
  - In one period, the stock price will either rise by $10 or fall by $10.
  - The one-period risk-free rate is 6%.

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# A Two-State Single-Period Model

- The payoffs can be summarized in a binomial tree

| 0 | 1 | | | |
|---|---|---|---|---|
| | *Stock* | *Bond* | *Call* | |
| **Stock** 50 | 60 | 1.06 | $max(60 - 50, 0) = 10$ | up state |
| **Bond** 1 | 40 | 1.06 | $max(40 - 50, 0) = 0$ | down state |

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# A Two-State Single-Period Model

- Let $\Delta$ be the number of shares of stock purchased,
- and let $B$ be the initial investment in bonds.
- To create a call option using the stock and the bond, the value of the portfolio consisting of the stock and bond must match the value of the option in every possible state.

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# A Two-State Single-Period Model

- In the up state, the value of the portfolio must be \$10.

$$60\Delta + 1.06B = 10$$

- In the down state, the value of the portfolio must be \$0.

$$40\Delta + 1.06B = 0$$

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# A Two-State Single-Period Model

- Using simultaneous equations, $\Delta$ and B can be solved for.
  - $\Delta = 0.5$
  - $B = -18.8679$
- A portfolio that is long 0.5 share of stock and short approximately \$18.87 worth of bonds will have a value in one period that exactly matches the value of the call.

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# A Two-State Single-Period Model

- By the Law of One Price, the price of the call option today must equal the current market value of the replicating portfolio.
- The value of the portfolio today is the value of 0.5 shares at the current share price of $50, less the amount borrowed.

$$50\Delta + B = 50(0.5) - 18.87 = 6.13$$

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# A Two-State Single-Period Model

- Note that by using the Law of One Price, we are able to solve for the price of the option without knowing the probabilities of the states in the binomial tree.

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# Replicating an Option in the Binomial Model

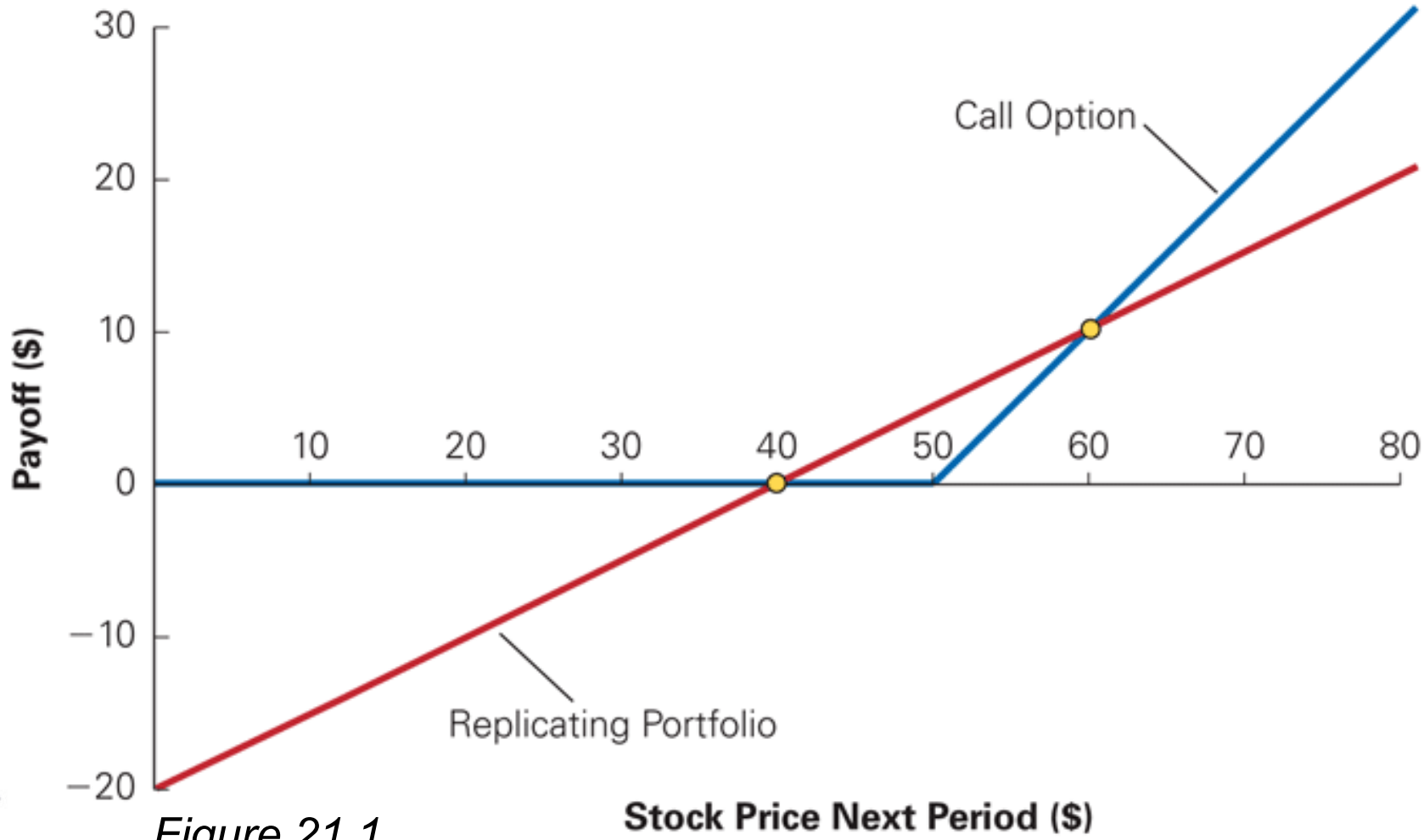


*Figure 21.1*

# The Binomial Pricing Formula

- Assume
  - $S$ is the current stock price, and $S$ will either go up to $S_u$ or go down to $S_d$ next period.
  - The risk-free interest rate is $r_f$.
  - $C_u$ is the value of the call option if the stock goes up and $C_d$ is the value of the call option if the stock goes down.

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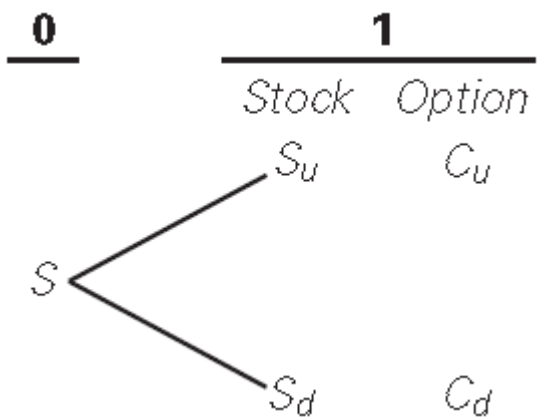
# The Binomial Pricing Formula

- Given the above assumptions the binomial tree would look like




- The payoffs of the replicating portfolio could be written as

$$S_u\Delta + (1 + r_f)B = C_u \text{ and } S_d\Delta + (1 + r_f)B = C_d$$

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# The Binomial Pricing Formula

- Solving the two replicating portfolio equations for the two unknowns $\Delta$ and B yields the general formula for the replicating formula in the binomial model.
  - Replicating Portfolio in the Binomial Model

$$\Delta = \frac{C_u - C_d}{S_u - S_d} \text{ and } B = \frac{C_d - S_d \Delta}{1 + r_f}$$

- The value of the option is:
  - Option Price in the Binomial Model

$$C = S\Delta + B$$

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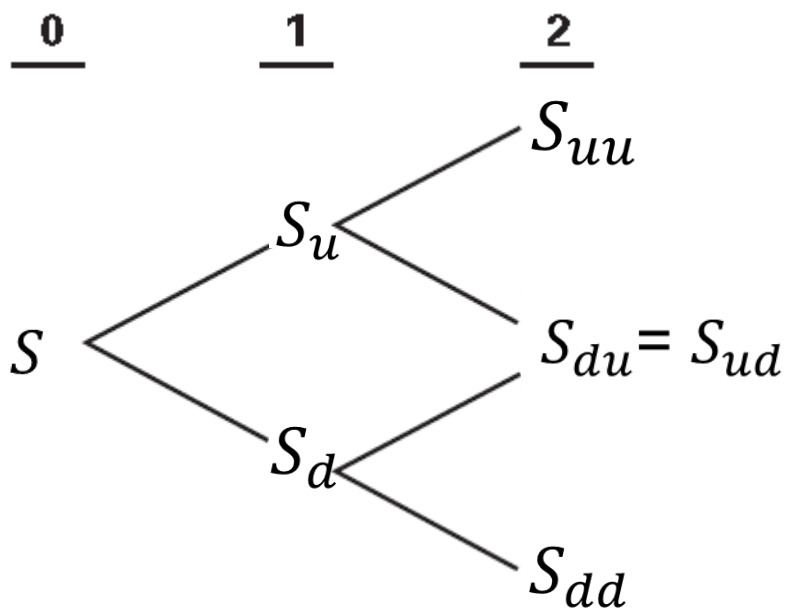
# A Multiperiod Model

- Consider a two-period binomial tree for the stock price.

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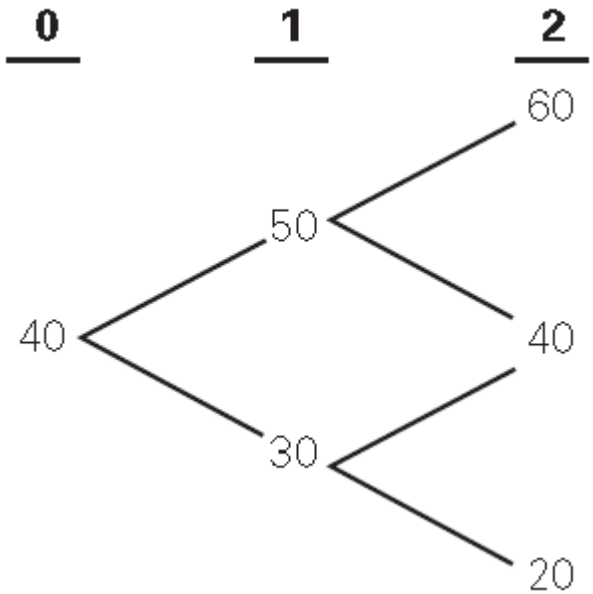
# A Multiperiod Model

- Dynamic Trading Strategy
  - A replication strategy based on the idea that an option payoff can be replicated by dynamically trading in a portfolio of the underling stock and a risk-free bond

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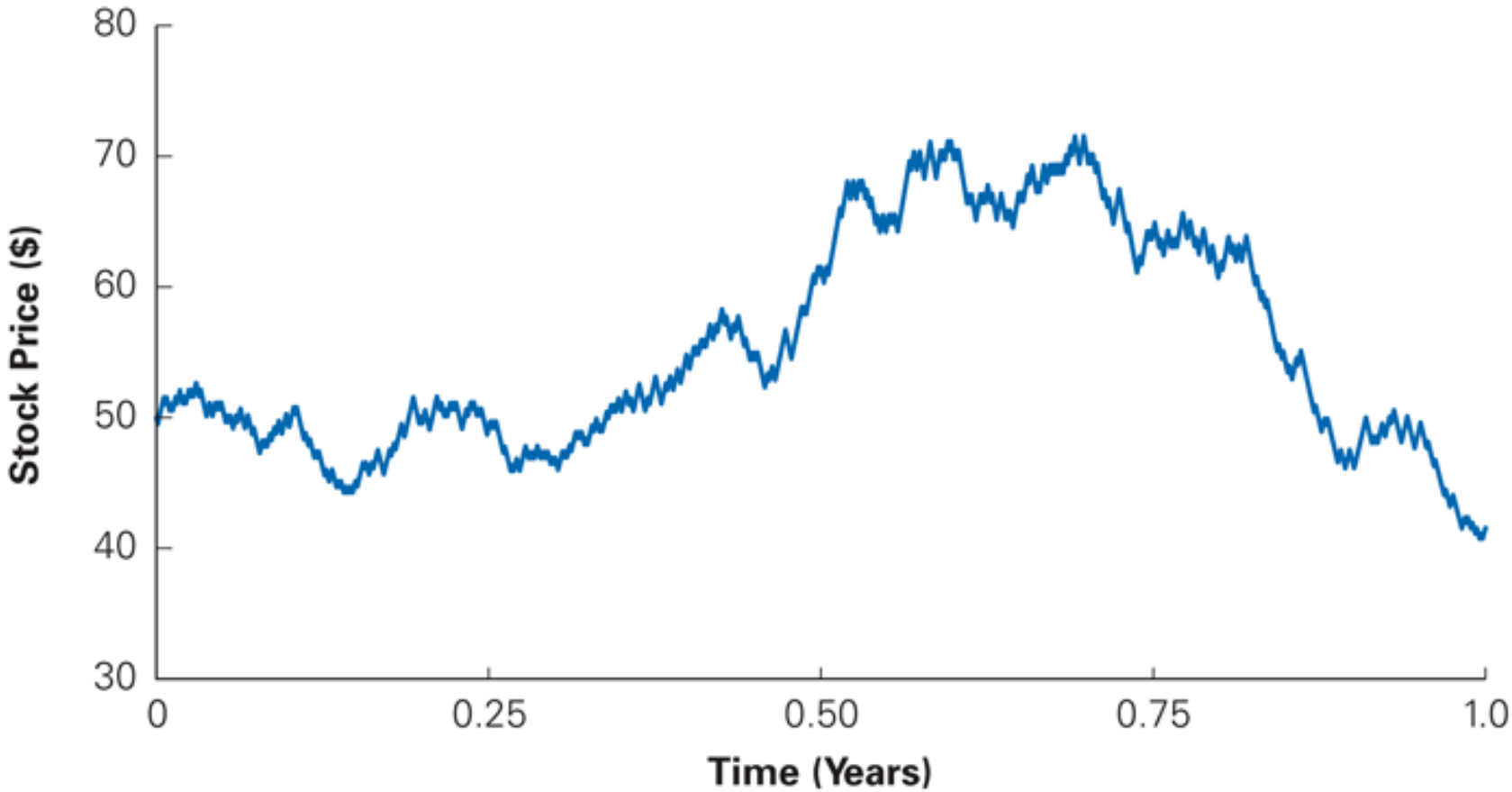
# Making the Model Realistic

- Although binary up or down movements are not the way stock prices behave on an annual or even daily basis, by decreasing the length of each period, and increasing the number of periods in the stock price tree, a realistic model for the stock price can be constructed.

# A Binomial Stock Price Path



# The Black-Scholes Option Pricing Model

- Black-Scholes Option Pricing Model
  - A technique for pricing European-style options when the stock can be traded continuously.
  - It can be derived from the Binomial Option Pricing Model by allowing the length of each period to shrink to zero and letting the number of periods grow infinitely large.

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# The Black-Scholes Formula

- Black-Scholes Price of a Call Option on a Non-Dividend-Paying Stock:

$$C = S \times N(d_1) - PV(K) \times N(d_2) \qquad (21.7)$$

  - Where S is the current price of the stock, K is the exercise price, and N(d) is the cumulative normal distribution
  - Cumulative Normal Distribution is the probability that an outcome from a standard normal distribution will be below a certain value

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# Normal Distribution

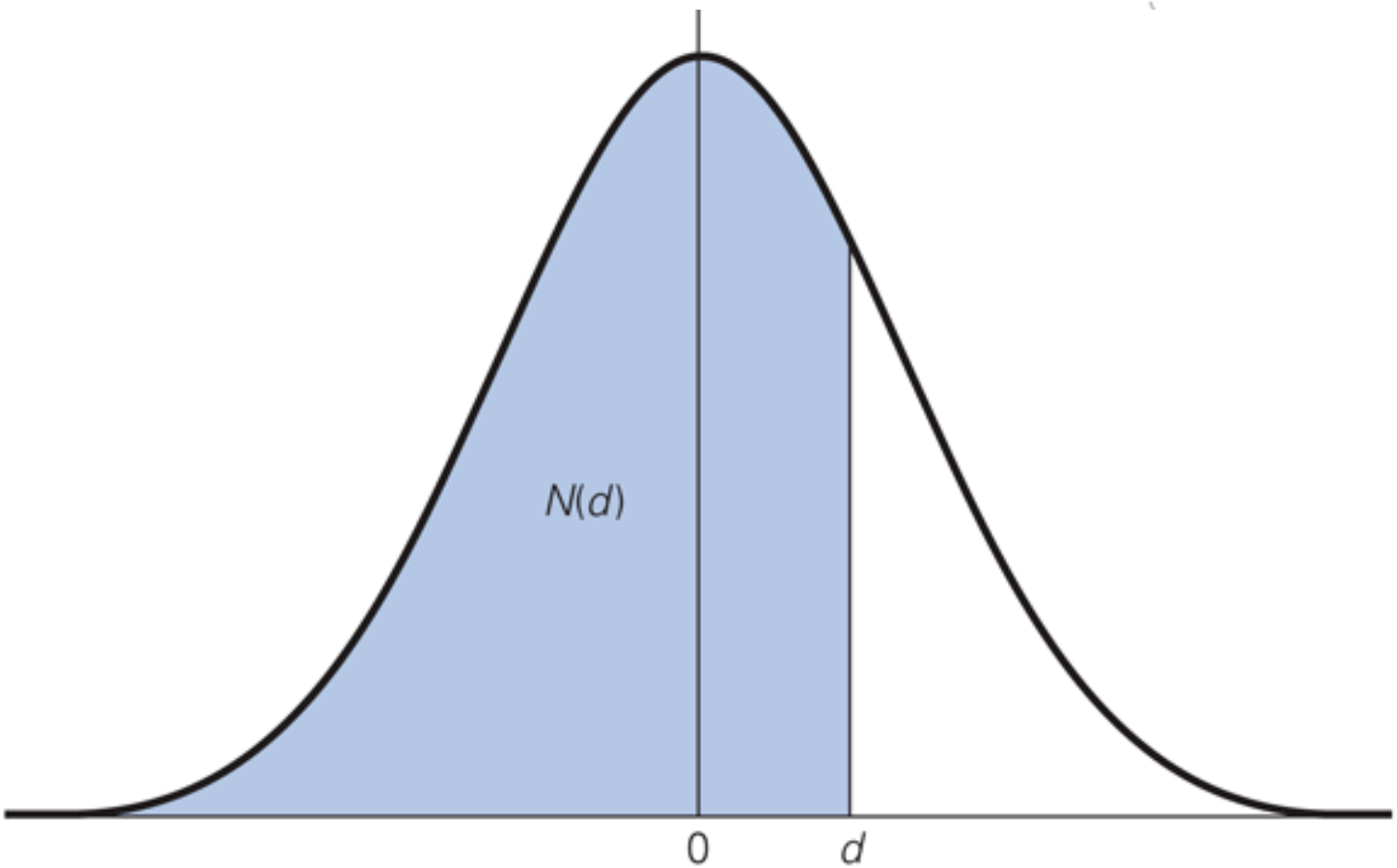


*Figure 21.3*

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# The Black-Scholes Formula

$$d_1 = \frac{\ln[S / PV(K)]}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} \quad and \quad d_2 = d_1 - \sigma\sqrt{T} \qquad (21.8)$$

- Where $\sigma$ is the annual volatility, and T is the number of years left to expiration

- Note: Only five inputs are needed for the Black-Scholes formula.
  - Stock price
  - Strike price
  - Exercise date
  - Risk-free rate
  - Volatility of the stock

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# Example

- Assume:
  - CLW Inc. does not pay dividends.
  - The standard deviation of CLW is 45% per year.
  - The risk-free rate is 5%.
  - CLW stock has a current price of $24.
- Using the Black-Scholes formula, what is the price for a ½ year American call option on CLW with a strike price of $30?

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Table 1: Table of the Standard Normal Cumulative Distribution Function $\Phi(z)$

| z | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| -3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| -3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| -3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| -3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| -3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| -2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| -2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| -2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| -2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| -2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| -2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| -2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| -2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| -2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| -2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| -1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| -1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| -1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| -1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| -1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| -1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| -1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| -1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| -1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| -1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| -0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| -0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| -0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| -0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| -0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| -0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| -0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |

# Example

$$PV(K) = \frac{\$30}{1.05^{1/2}} = \$29.277$$

$$d_1 = \frac{\ln[\$24 / \$29.277]}{.45\sqrt{.5}} + \frac{.45\sqrt{.5}}{2} = -0.4658$$

$$d_2 = -0.4658 - .45\sqrt{.5} = -0.7840$$

$$N(-0.4658) = 0.3207 \qquad N(-0.7480) = 0.2165$$

$$C = \$24 \times 0.3207 - \$29.277 \times 0.2165 = \$1.36$$

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# European Put Options

- Using put-call parity, the value of a European put option is $P = C - S + PV(K)$
- Black-Scholes Price of a European Put Option on a Non-Dividend-Paying Stock

$$P = PV(K)[1 - N(d_2)] - S[1 - N(d_1)] \qquad (21.9)$$

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# Summary

- **The Binomial Option Pricing Model**
  - An option can be valued using a portfolio that replicates the payoffs of the option in different states.
  - The Binomial Option Pricing Model assumes two possible states for the next time period, given today's state.
  - The value of an option is the value of the portfolio that replicates its payoffs. The replicating portfolio will hold the underlying asset and risk-free debt, and will need to be rebalanced over time.
- The replicating portfolio for the Binomial Option Pricing Model is

$$\Delta = \frac{C_u - C_d}{S_u - S_d} \text{ and } B = \frac{C_d - S_d\Delta}{1 + r_f} \quad (21.5)$$

- Given the replicating portfolio, the value of the option is

$$C = S\Delta + B \quad (21.6)$$

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# Summary

**The Black-Scholes formula**

- The Black-Scholes option pricing formula for the price of a call option on a non dividend-paying stock

$$C = S * N(d_1) - PV(K) * N(d_2) \qquad (21.7)$$

- where $N(d)$ is the cumulative normal distribution and

$$d_1 = \frac{\ln[S / PV(K)]}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} \quad and \quad d_2 = d_1 - \sigma\sqrt{T}$$

- The Black-Scholes formula for the price of a European put option on a non-dividend paying stock is

$$P = PV(K)[1 - N(d_2)] - S[1 - N(d_1)]$$

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